COURSEWORK ASSIGNMENT SUBMISSION

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Subject: Software Engineering & Project Management **Day:** Day 29 — Project Review, Quality Audit & Final Demonstration

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Section 9: Classroom Activity — Run a Mock Project Review

Formal project quality review and stage-gate audit for the College Event Management System (CEMS) Release 1.0.

1. Requirements-to-Evidence Verification Checklist

Requirement ID	Requirement Description	Verification Evidence Provided	Audit Status
REQ-01	Institutional Google SSO Authentication	Cypress automated auth test log & OAuth trace.	VERIFIED (Pass)
REQ-02	Real-time Event Catalog Search & Filters	UI Lighthouse report & Jest component tests.	VERIFIED (Pass)
REQ-03	Atomic Seat Allocation & Concurrency Control	k6 concurrent load test report (0 race conditions).	VERIFIED (Pass)
REQ-04	Digital QR Ticket Generation	Mobile screenshot & QR barcode scanner decode log.	VERIFIED (Pass)
REQ-05	Gate Check-in & Attendance Marking	Gate scanner integration test; database update log.	VERIFIED (Pass)

2. Defect Triage & Severity Matrix

Defect ID	Defect Summary	Severity	Priority	Resolution Action
BUG-101	Special characters in student names cause PDF badge render failure.	Major	High	Sanitize string inputs; update PDF font library.
BUG-102	Mobile footer overlaps register button on iPhone SE (320px width).	Minor	Medium	Add CSS bottom padding media query.
BUG-103	Admin CSV export includes deleted test event records.	Major	High	Add `WHERE is_deleted = false` filter to query.

3. Formal Release Recommendation

Verdict: READY WITH ACCEPTED LIMITATIONS.
Justification: All 5 core functional requirements are 100% verified. Zero Critical severity defects remain open. The two Major defects (BUG-101 name sanitization and BUG-103 CSV filter) have verified patches queued for staging deployment. Mobile layout issue BUG-102 is accepted as a minor cosmetic limitation affecting < 2% of user screens, with a fix scheduled in the first maintenance patch.

Section 12: Homework — Quality Review & Release Recommendation

Formal quality audit report, five-factor risk register, and demonstration script for CEMS.

1. Five-Risk Operational Register

Risk Description	Probability	Impact	Planned Operational Response	Risk Owner
Live Wi-Fi drops at campus auditorium gates.	High	Medium	Enable offline PWA caching on scanner tablets.	Lead Tester
Sudden registration rush crashes web server.	Medium	High	Configure Nginx rate limiting and AWS auto-scaling.	DevOps Lead
Faculty approver unavailable for event release.	Medium	Medium	Implement automated approval delegation fallback.	Project Lead
SMTP server hits outbound email rate limit.	Low	High	Route emails through verified SendGrid relay service.	Backend Dev
Student inputs invalid mobile phone number.	High	Low	Implement client-side regex phone number validation.	Frontend Dev

2. Five-Minute Live Demonstration Script & Fallback Plan

- **Minute 0:00–1:00:** Introduction, project objectives, and architecture overview.
- **Minute 1:00–2:30:** Live walkthrough: Student logs in via SSO, searches for "Robotics Hackathon", registers, and displays dynamic QR confirmation ticket.
- **Minute 2:30–3:30:** Gate check-in demo: Volunteer scans student QR code on mobile camera; attendance updates in real time.
- **Minute 3:30–4:30:** Coordinator view: Real-time dashboard seat counter update and CSV attendance sheet download.
- **Minute 4:30–5:00:** Technical summary, automated test execution metrics, and Q&A.
- **Demonstration Fallback Plan:** If live campus internet drops during the presentation: (1) Switch immediately to pre-configured local Docker container environment running on `localhost:3000`; (2)

If hardware fails, execute demonstration using a recorded 1080p high-resolution screen walkthrough video backed by pre-exported test logs.

DAKSH